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Peptacular: A ProForma 2.1 compliant Python package for amino acid sequence analysis

Patrick T. Garrett [1], **Titus Jung**[1], **and John R. Yates III** [1¶]

1 The Scripps Research Institute, United States ¶ Corresponding author

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Summary

Mass spectrometry-based proteomics depends on computational methods to identify and characterize (amino acid) AA sequences. These sequences, ranging from short peptides to complete proteins, exhibit substantial chemical complexity due to PTMs, variable charge states, neutral losses, and isotopic patterns (Angel et al., 2012; Smith & Kelleher, 2013). **Peptacular** is a Python library designed to handle this complexity. The library provides functionality for modifying, calculating mass, m/z, isotopic distributions, physicochemical properties, enzymatic digestion, and fragmentation of AA sequences. Built around the standardized ProForma 2.1 notation, it supports all non-cross-linked ProForma features.

Statement of Need

Historically, the proteomics field has lacked standardization for representing AA sequences. Individual software tools have implemented proprietary notations, which has created barriers to data integration and reanalysis across platforms. ProForma notation (LeDuc et al., 2018) was developed to address this challenge by providing a unified representation system. However, adoption has remained limited, partly due to insufficient support in widely-used computational tools and libraries. Peptacular was specifically designed to accelerate ProForma adoption by offering a comprehensive and accessible API with clear documentation.

Additionally, with the continued advance of mass spectrometers, modern proteomics experiments routinely identify tens of thousands of AA sequences. These results are typically exported as tabular files and are frequently processed using Python, particularly with data manipulation libraries such as pandas (Team, 2025) and polars (Vink et al., 2025). To support this workflow, Peptacular's functional API allows you to directly manipulate these tabular data structures and automatically parallelizes operations, making large-scale sequence analysis both fast and straightforward (See example 2).

State of the Field

Several Python packages provide AA sequence analysis capabilities, though each has limitations as a general-purpose library for ProForma notation. **Pyteomics** (Goloborodko et al., 2013) recently added partial ProForma support while maintaining its legacy notation system, requiring format conversions that are not universally supported. **BioPython** (Cock et al., 2009) offers limited support for AA sequences and has no ProForma compatibility; its broad scope includes DNA and RNA sequences. **RustyMS** (Schulte et al., n.d.) delivers comprehensive ProForma parsing through its Python bindings but focuses primarily on parsing and offers limited prebuilt functionality for working with sequences. **PyOpenMS** (Röst et al., 2013) provides an extensive mass spectrometry toolkit but lacks ProForma compatibility.

Peptacular was designed to address these limitations as a general-purpose, ProForma 2.1-compliant AA sequence library. Developing Peptacular from scratch rather than extending existing tools offered three key advantages. First, ProForma notation serves as the foundation rather than a retrofitted addition, enabling complete feature coverage and a cleaner API. Second, the library targets AA sequence analysis specifically, avoiding the complexity that arises from supporting a broader scope. Third, the pure Python implementation ensures compatibility with modern Python versions, including free-threaded builds with the Global Interpreter Lock (GIL) disabled, which will become increasingly relevant as Python's parallelization capabilities continue to improve.

Software Design

Peptacular provides two primary APIs: a functional API and an object-oriented API. The object-oriented API employs a factory pattern to modify Proforma Annotation objects, enabling precise control over annotations. The functional API provides functions that operate directly on serialized sequences and annotations, with automatic parallelization for batch processing.

The built-in parallelization supports three execution backends: sequential, threaded, and process-based (default). Sequential execution provides single-threaded processing for small batches where parallelization overhead is detrimental. Thread-based parallelization currently offers limited benefits due to the GIL but will improve as free-threaded Python builds become standard (PEP 703). Process-based parallelization is the default and most widely supported method. Process and thread spawning mechanisms (fork, spawn, forkserver) are globally configurable, and worker processes are cached to eliminate startup overhead.

Three performance optimizations enable efficient large-scale processing. First, lazy evaluation keeps modifications in serialized form until required. Second, aggressive caching exploits the fact that proteomics datasets typically contain a small number of repeated modifications. Third, the specific objects used to store modifications within the annotations objects are only initialized when needed, reducing memory footprint and accelerating object creation.

Modification reference data, including masses, compositions, and identifiers from Unimod (Creasy & Cottrell, 2004), PSI-MOD (Hupo-Psi, n.d.-a), RESID (*RESID Database [PIR - Protein Information Resource]*, n.d.), XLMOD (Hupo-Psi, n.d.-b), and GNOme (Glygen-Glycan-Data, n.d.), are provided by the companion package **Tacular** (P. Garrett, 2026). This package embeds the data directly within itself as Python modules rather than storing them as external files. Only valid modifications are included in the embedded data; a modification is considered valid if it possesses at least one of the following properties: average mass, monoisotopic mass, or chemical formula. This design eliminates file I/O overhead during the parsing of supported ontologies.

The package includes full type annotations with a py.typed marker, which enables static type checking and provides IDE autocomplete, inline documentation, and compile-time error detection. Test coverage exceeds 70%, with continuous integration implemented through GitHub Actions. The only dependency is the companion package: tacular.

Research impact statement

Since its initial release, Peptacular has demonstrated measurable adoption. The package has accumulated over **33k** downloads from PyPI (as of 07 February 2026), with sustained weekly download rates exceeding 200 installations. It has been listed as one of three Python packages to support ProForma notation by the PSI group. Additionally, Peptacular was used to generate figures within a textbook chapter (P. T. Garrett et al., 2025).

Example Usage

Object-Based API

```python
import peptacular as pt

# Parse a sequence into a ProFormaAnnotation
peptide: pt.ProFormaAnnotation = pt.parse("PEM[Oxidation]TIDE")

# Calculate mass and m/z
mass: float = peptide.mass() # 849.342
mz: float = peptide.mz(charge=2) # 425.678

# Factory pattern
print(peptide.set_charge(2).set_peptide_name("Peptacular").serialize())
# (>Peptacular)PEM[Oxidation]TIDE/2
```

Functional-Based API

```python
import peptacular as pt

peptides = ['[Acetyl]-PEPTIDES', '<C13>ARE', 'SICK/2']

# Calculate mass and m/z for all peptides
masses: list[float] = pt.mass(peptides) # [928.4026, 374.1914, 451.2454]
mzs: list[float] = pt.mz(peptides, charge=2) # [465.2086, 188.103, 225.6227]
```

Pandas-Functional API

```python
import peptacular as pt
import pandas as pd

df = pd.DataFrame(
    {
        "seq": ["PEM[Oxidation]TIDE", "ACDEFGHIK", "M[Phospho]NOPQR"],
    }
)

df["mass"] = df["seq"].apply(pt.mass)
```

Mathematics

Peptacular calculates the molecular mass and isotopic patterns for AA sequences. This section presents the mathematical framework that underlies these calculations.

Base Mass

The base mass M_{base} of a AA sequence containing modifications is calculated as the sum of all constituent molecular components:

$$M_{base} = \sum_{i=1}^{n} m_{AAi} + M_N + M_C + M_S + M_I + M_R + M_U + \mathbb{1}_{\text{precursor}} \cdot M_L$$

where:

- n is the sequence length
- m_{AA_i} is the mass of amino acid at position i

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- M_N is the total mass of N-terminal modifications
- M_C is the total mass of C-terminal modifications
- M_S is the total mass of static/fixed modifications (Applied to sequence)
- M_I is the total mass of position-specific modifications
- M_R is the total mass of modifications within defined sequence intervals
- M_U is the total mass of modifications with unknown positions
- M_L is the total mass of labile modifications
- $\mathbb{1}_{\text{precursor}}$ is an indicator function: 1 for precursor ions, 0 for fragment ions

Labile modifications are included only for precursor and neutral ion types.

Neutral Mass

The neutral mass M_{neutral} is calculated by combining the base mass with ion-type adjustments, isotope modifications, neutral deltas.

$$M_{\text{neutral}} = M_{\text{base}} + M_{\text{ion}} + M_{\text{isotope}} + M_{\text{ndelta}}$$

where:

- M_{base} is the peptide base mass from the previous section
- M_{ion} is the ion-type-specific mass offset
- M_{isotope} is the mass shift from a specific isotopic species
- M_{ndelta} is the mass change from neutral losses/gains

Mass-to-charge Ratio

The mass-to-charge (m/z) ratio is calculated by incorporating charge carriers and electron mass corrections to the neutral mass:

$$\frac{m}{z} = \frac{M_{\text{neutral}} + M_{\text{adduct}} - z \cdot m_e}{z}$$

where:

- M_{neutral} is the neutral fragment mass
- M_{adduct} is the total mass of charge carriers
- z is the total charge state
- $m_e = 0.0005485799$ Da (electron mass)

Figures

Table 1: Proforma 2.1 Compliance

?	Feature	Example	§ [Support]
Y	Amino acids (+UO)	`AAHCFKUOT`	6.1 [B]
Y	Unimod names	`PEM[Oxidation]AT`	6.2.1 [B]
Y	PSI-MOD names	`PEM[monohydroxylated residue]AT`	6.2.1 [B]
Y	Unimod numbers	`PEM[UNIMOD:35]AT`	6.2.2 [B]
Y	PSI-MOD numbers	`PEM[MOD:00425]AT`	6.2.2 [B]
Y	Delta masses	`PEM[+15.995]AT`	6.2.3 [B]
Y	N-terminal modifications	`[Carbamyl]-QPEPTIDE`	6.3 [B]
Y	C-terminal modifications	`PEPTIDEG-[Methyl]`	6.3 [B]
Y	Labile modifications	`{Glycan:Hex}EM[U:Oxidation]EV`	6.4 [B]
Y	Multiple modifications	`MPGNW[Oxidation][Carboxymethyl]PESQE`	6.5 [B]
Y	Information tag	`ELV[INFO:AnyString]IS`	6.6 [B]
Y	Ambiguous amino acids	`BZJX`	7.1 [2]
Y	Prefixed delta masses	`PEM[U:+15.995]AT`	7.2 [2]

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?	Feature	Example	§ [Support]
Y	Mass gap	`PEX[+147.035]AT`	7.3 [2]
Y	Formulas	`PEM[Formula:O]AT`, `PEM[Formula:[1701]]AT`	7.4 [2]
Y	Mass with interpretation	`PEM[+15.995|Oxidation]AT`	7.5 [2]
Y	Unknown mod position	`[Oxidation]?PEMAT`	7.6.1 [2]
Y	Set of positions	`PEP[Oxidation#1]M[#1]AT`	7.6.2 [2]
Y	Range of positions	`PRT(ESFRMS)[+19.0523]ISK`	7.6.3 [2]
Y	Position scores	`PEP[Oxidation#1(0.95)]M[#1(0.05)]AT`	7.6.4 [2]
Y	Range position scores	`(PEP)[Oxidation#1(0.95)]M[#1(0.05)]AT`	7.6.5 [2]
Y	Amino acid ambiguity	`(?VCH)AT`	7.7 [2]
Y	Modification prefixes	`PEPM[U:Oxidation]AS[M:O-phospho-L-serine]`	7.8 [2]
Y	RESID modifications	`EM[R:L-methionine sulfone]EM[RESID:AA0581]`	8.1 [T]
Y	Names	`(>Heavy chain)EVQLVESG`	8.2 [T]
Y	XL-MOD modifications	`EVTK[X:Aryl azide]LEK[XLMOD:00114]SEFD`	9.1 [X]
N	Cross-linkers (intrachain)	`EVTK[X:Aryl azide#XL1]LEK[#XL1]SEFD`	9.2.1 [X]
N	Cross-linkers (interchain)	`EVTK[X:Aryl azide#XL1]L//EK[#XL1]SEFD`	9.2.2 [X]
N	Branches	`ED[MOD:00093#BRANCH]//D[#BRANCH]ATR`	9.3 [X]
Y	GNO modifications	`NEEYN[GNO:G59626AS]K`	10.1 [G]
Y	Glycan compositions	`NEEYN[Glycan:Hex5HexNAc4NeuAc1]K`	10.2 [G]
Y	Charged formulas	`SEQUEN[Formula:Zn1:z+2]CE`	11.1 [A]
Y	Controlling placement	`PTI(MERMERME)[+32|Position:E]PTIDE`	11.2 [A]
Y	Global isotope	`<13C>CARBON`	11.3.1 [A]
Y	Fixed modifications	`<[Oxidation]@M>ATPEMILTCMGCLK`	11.3.2 [A]
Y	Chimeric spectra	`NEEYN+SEQUEN`	11.4 [A]
Y	Charges	`SEQUEN/2`, `SEQUEN/[Na:z+1,H:z+1]`	11.5 [A]
Y	Ion notation	`SEQUEN-[b-type-ion]`	11.6 [A]

Table 1 presents the level of ProForma support implemented in Peptacular. The package currently supports all ProForma 2.1 features for linear peptides. Cross-linked peptides (both inter- and intrachain) and branched structures are not currently supported. Ion notation is also not supported at the sequence level; however, the package provides extensive fragmentation support through either API. Support levels are designated as follows: [B] - Base ProForma support, [2] - ProForma 2, [T] - Top down, [X] - Cross linking, [G] - Glycan, [A] - Advanced.

AI usage disclosure

Generative AI models were employed to support the development of this software package. Specifically, Claude Code, Cursor, and GitHub Copilot were utilized for code generation, test development, debugging assistance, and documentation preparation. Additionally, Type.ai was used to assist in manuscript preparation. All AI-generated content was subsequently reviewed and verified for accuracy.

Availability

Peptacular is distributed through PyPI (https://pypi.org/project/peptacular/) and available as open-source software on GitHub (https://github.com/tacular-omics/peptacular). Documentation is accessible at https://peptacular.readthedocs.io. The software is released under the MIT license.

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